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MÁSTER UNIVERSITARIO EN SISTEMAS ESPACIALES

Work Breakdown Structure and Gantt chart

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1 Introduction

This document explains in detail two fundamental elements in the management of the Blue Horizon Mission project: the Gantt Chart, where all the project phases and milestones will be organised temporally, and the Work Breakdown Structure, where all the tasks necessary to carry out the project will be organised by subsystems;

2 Gantt Chart

The main target of this diagram is to minimise the project development time as much as possible without penalising the quality of the project. Below is the Gantt chart of the project divided by phases, and a brief summary of each of these tasks is given. The duration times for each process have been chosen based on typical mission duration times for cubesats of similar size to ours.

2.1 Phase 0/A

- **Kick-off meeting and identify needs** First meeting of the project in which all team members participate. In this meeting, the objectives and characteristics of the project are presented and internalised.
- **Elaborate mission statement:** Phase of the project in which the objective and purpose of the mission is developed and defined in detail.
- **Develop preliminary technical requirements:** During this phase, a first iteration of the technical requirements of the mission is established, distinguishing the high-level requirements and those of each subsystem, drawing up an SRD that will be updated as the mission develops.
- **Propose mission concepts:** Stage in which a preliminary mission concepts is developed and presented. This concept includes initial ideas on how the mission will be carried out, what technologies and methods will be used.
- **Preliminary study of risks:** Early stage of the project in which potential risks associated with the mission are identified and analysed.
- **MDR (Mission Definition Review):** Important milestone in which the status of mission concepts are assessed.
- **MDR Resolution:** Following the previous milestone, a new phase begins focused on providing recommendations to refine the mission definition and address the identified issues, as well as establishing the final resolution of this process.
- **Propose systems concepts and architectures:** Crucial phase in which the general ideas and concepts of how the different systems will work and how they will be configured in the structure are developed and presented.

- **Develop preliminary management plan:** Initial document establishing the basis for the management and organization of the project. Here is included the creation of the WBS and the product tree.
- **Develop preliminary systems engineering plan:** This plan defines the strategy and approach that will guide the design and integration of all systems involved in the mission and ensures that systems engineering principles are followed throughout the project life cycle. Here is included the creation of the Gantt Chart.
- **Develop preliminary product assurance plan:** Key document in the development of a space mission that sets out the strategies and actions needed to ensure that products (systems, components, and subsystems) meet quality, reliability, safety and performance requirements throughout the project life cycle.
- **Study technical and programmatic feasibility:** It is self-explanatory, once the preliminary requirements and their organisation have been proposed, the feasibility of this first iteration should be studied.
- **Identify critical technologies and elements:** This phase involves identifying and assessing the essential technologies and components that are necessary for mission success, including recognising the vital components and its TRL that must be designed, manufactured and tested.
- **Update risk assessment:** Once the critical technologies have been studied, it is necessary to reassess the risks of these technologies and how they affect the mission.
- **Preliminary cost assessment:** Study the costs of the initial configuration, including possible facilities, suppliers and launchers.
- **Design and selection of available options for phase B:** Different options that are feasible in the first instance are designed for further study during phase B.
- **PRR (Preliminary Requirements Review):** Crucial milestone before finishing this phase where the preliminary requirements of the project are reviewed and validated. This is a comprehensive assessment to ensure that the project objectives and specifications are clearly defined and feasible.

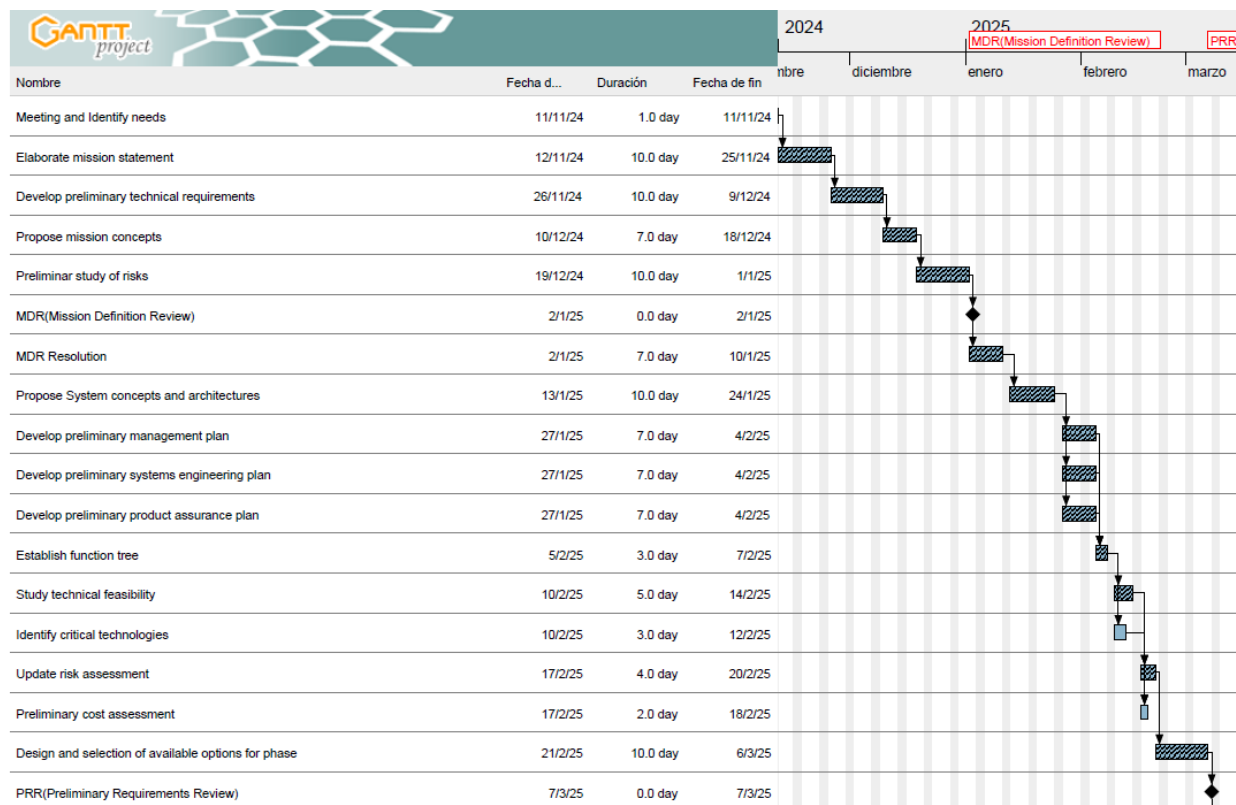


Figure 2.1: Phase OA

2.2 Phase B

- **PRR resolution:** Selection of the most viable design concepts that best suit to the specified requirements.
- **Launcher selection:** Selection of various launch vehicles, visualization of flight schedules, evaluation of technical requirements and cost estimation for each flight, and selection of the final launchers (two launchers, as mission requires).
- **Prepare providers business agreements:** Meetings and agreements with COTS components providers for the satellites.
- **Finish management, systems engineering and assurance plans:** To ensure that the project develops correctly.
- **Elaborate baseline costs at completion:** To evaluate the total project cost and identify the areas where the highest expenditures occur.
- **Preliminary OBS:** Assignment of the personnel in charge of each work package (WBS predefined in preliminary management plan).
- **Confirm feasibility of the technical solutions of the system and operations concept:** To determine if design changes are necessary to reduce the risks of the mission.

- **Trade off studies and selection of preferred system concept:** Meetings with the customer in order to see which requirements must be more flexible and select the preferred design option.
- **Preliminary design definition of BlueHorizon:** Once the design option is selected, a first iteration for the design of the system must be done.
- **Prepare customers business agreements:** Negotiation, redaction and formalization of terms and conditions of the agreements between both sides. It will last until phase C beginning.
- **Preliminary verification program:** Verify that the design completed so far meets the specified requirements.
- **Preliminary definition of external interfaces:** Check the interfaces between satellites and launcher, and also between satellites and their payload.
- **Systems Requirements Review:** Meeting to check that requirements are met, viable and aligned with mission objectives.
- **Initiate long-lead item procurement:** Identification of the long-lead items required, search of providers, negotiations, formalization of agreements with providers, and organization of the reception of each component within its delivery time.
- **Begin space debris mitigation plan:** Preliminary definition of possible actions to be made to prevent the formation of space debris.
- **Begin disposal plan:** Preliminary definition of de-orbit strategies applicable to each satellite of the constellation after their functional time.
- **Begin reliability and safety assessment:** Initiation of activities that ensure system performance and risk mitigation.
- **Release of product tree:** When finished the agreements with provided, product tree must be completed and released.
- **Release of WBS:** Finalization of Work Breakdown Structure initiated in Phase 0/A (management plan).
- **Elaborate specification tree:** Detailed definition of the low-level requirements of each subsystem.
- **Update risk assessment:** Revision of risks to reflect current conditions of the project.
- **PDR (Preliminary Design Review):** Meeting to evaluate whether the system design is progressing according to the requirements and whether the proposed solution remains within the acceptable risks.

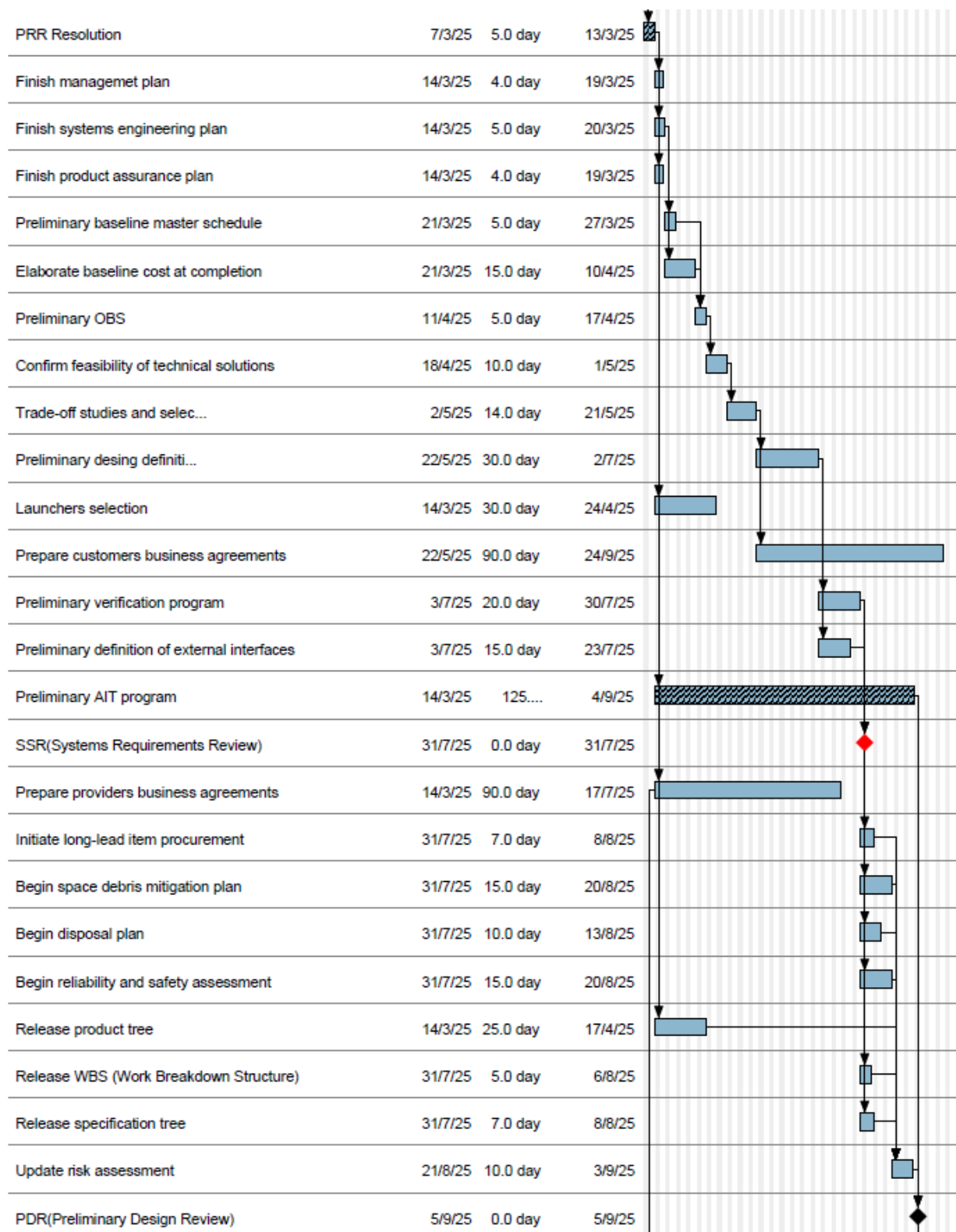


Figure 2.2: Phase B

2.3 Phase C

- **PDR Resolution:** In-depth feasibility study of the selected final design in order to start phase C.
- **Ground segment definition:** Study of the availability of the facilities necessary for the exchange of data with the ground.
- **Arrival of components and Payload:** Corresponds to the waiting time between purchase and acquisition of the components.
- **Required infrastructure assesment:** Assessment of the facilities needed to carry out the mission.
- **Complete detailed design of all subsystem:** Final design of each subsystem separately.
- **Engineering model development testing:** Tests performed to ensure that components and subsystems perform as expected. These tests include integration and functional tests.
- **Schedule negotiation for qualification and verification tests:** It involves discussions and agreements between different stakeholders (engineering teams, project management, suppliers, etc.) to establish an efficient and reliable schedule for the mentioned tests.
- **Transport of components to Montegancedo**
- **Complete integration, verification and qualification plannings:** Planning of the set of critical activities that are essential to ensure that mission systems, components and subsystems function properly and meet established requirements.
- **Detailed definition of internal and external interfaces:** Identification and precise definition of the interactions and connections between the different components of the mission, both internal and external, including the interaction between the different subsystems and between the satellite modules.
- **Issue preliminar user manual:** Key document explaining how to use, operate and maintain the satellites.
- **Update risk assesment**
- **CDR(Critical Design Review):** Full design review before finishing phase C to ensure that the detailed design of the mission and its systems is fully developed and that all mission and technical requirements have been met before proceeding to final construction or manufacture of components and systems.

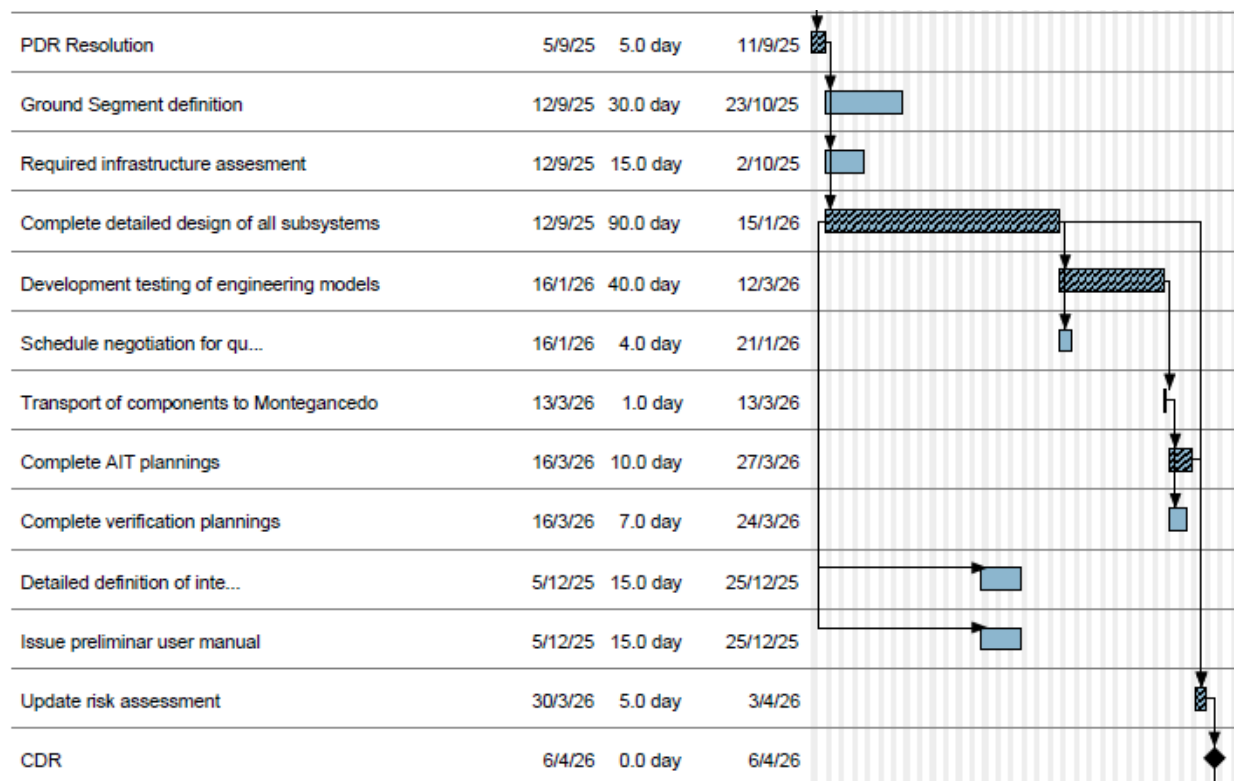


Figure 2.3: Phase C

2.4 Phase D

- **CDR Resolution**
- **Assembly of the qualification model:** The necessary components are assembled to build the model satellite for the qualification tests.
- **Functional testing of the qualification model**
- **Design In-flight calibration plan:** A plan is performed to control the calibration of the instruments of the satellites during the flight preparing the LEOP plan.
- **Complete interoperability tests:** The different subsystems of the satellite are verified for correct simultaneous operation.
- **Qualification test:** The necessary tests are performed for the qualification of the qualification model.
- **QR (Qualification Review):** Crucial milestone to ensure that the satellite constellation is able to reach the operational orbit safely.
- **Assembly and integration of the constellation:** At this time, the entire constellation is assembled.

- **Acceptance test:** This test is applied to verify that the entire process has been executed correctly.
- **Prepare acceptance data packages:** Corresponds to the complete and detailed documentation collected to demonstrate that the system meets the design and quality requirements prior to final acceptance.
- **Finalize preparation for operations and utilization:** Final preparations are performed prior to the acceptance test.
- **AR (Acceptance Review):** Crucial milestone to judge the readiness of the product for delivery. It is necessary that the constellation satellites conform to the initial qualification model.
- **ORR (Operational Readiness Review):** final milestone to move to phase E.

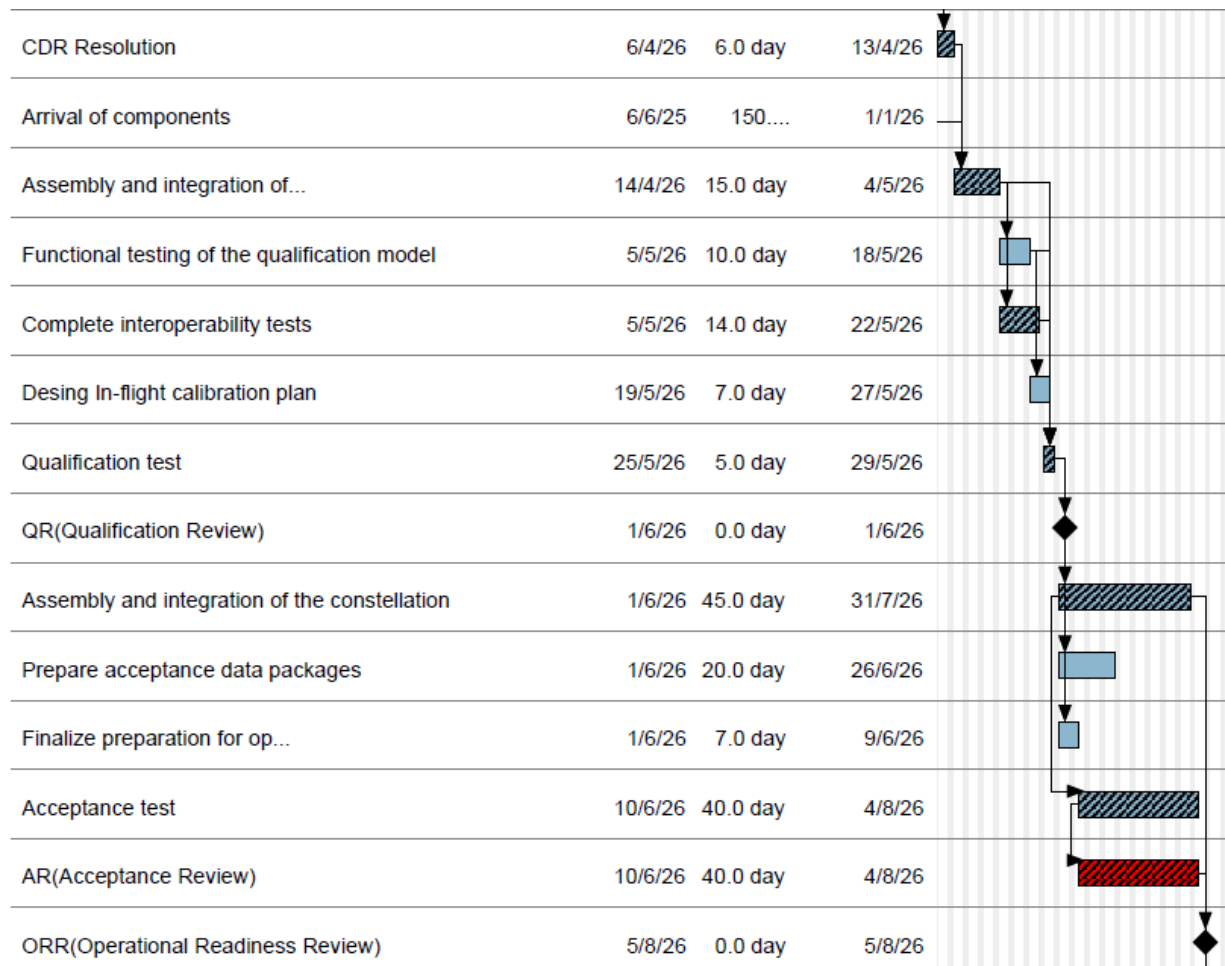


Figure 2.4: Phase D

2.5 Phase E

- **FRR (Flight Readiness Review):** Review carried out to ensure that all technical, operational and logistical aspects of the mission are in place for the successful launch and execution of the space mission.
- **Prepare launch campaign:** It involves all actions necessary to ensure that the launch of the mission is carried out successfully and as planned.
- **Transport half constellation to 1^o launch site**
- **Transport half constellation to 2^o launch site**
- **1^o spacecraft assembly to launcher**
- **2^o spacecraft assembly to launcher**
- **1^o LRR (Launch Readiness Review):** Ensures that all aspects of the mission are ready to proceed with the launch.
- **2^o LRR (Launch Readiness Review)**
- **1^o Launch**
- **2^o Launch**
- **1^o LEOP:** All the operations that occur between the launch and the nominal phase, such as the starting of the constellation, or the transition to the nominal orbits.
- **2^o LEOP**
- **Nominal Phase**
- **CRR (Commissioning Result Review):** commissioning to check that everything is working nominally a few days after the starting.
- **ELR (End of Life Review):** Verification that the systems are capable of performing the deorbiting as planned.

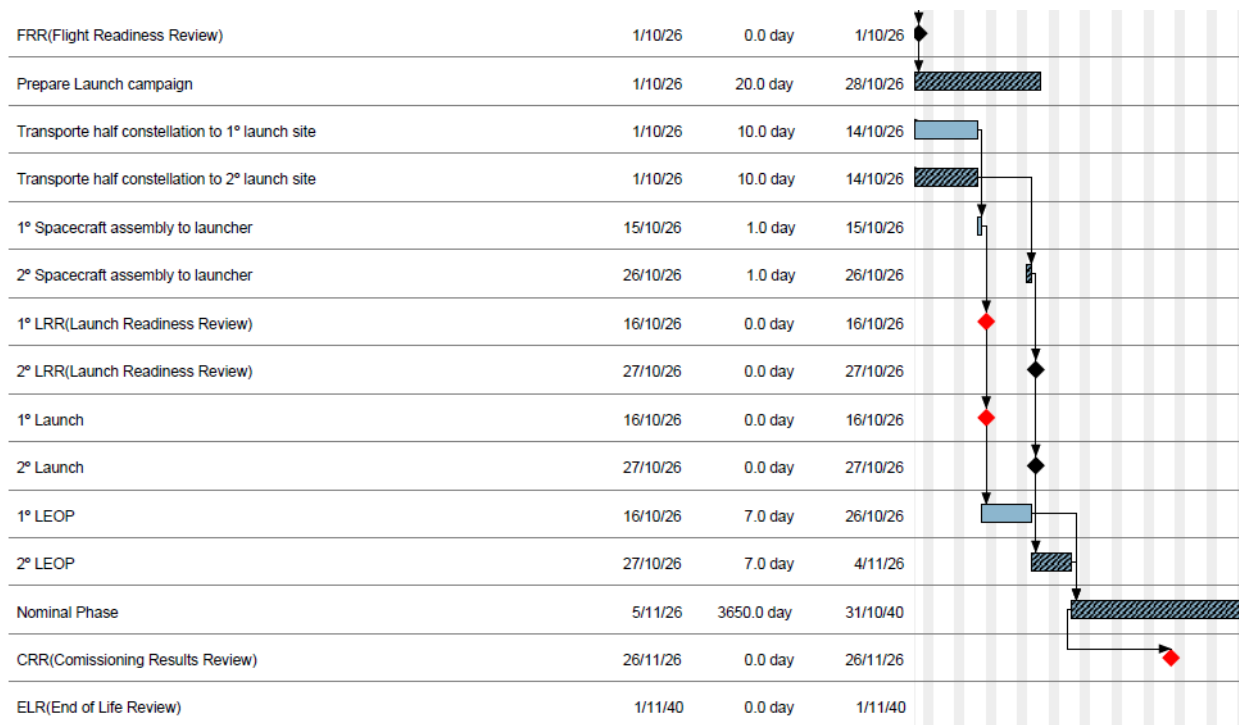


Figure 2.5: Phase E

2.6 Phase F

- Implement disposal plan
- **MCR (Mission Closed-out Review):** The MCR is a formal review that takes place at the end of the mission, assessing overall performance and verifying that all elements of the mission have been properly closed.

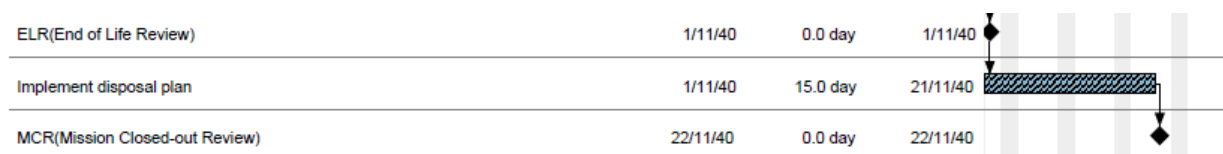


Figure 2.6: Phase F

3 Work Breakdown Structure

The WBS is a top-down hierarchical decomposition which allows to organize and classify the milestones of a project by breaking down the work to be done according to the order in which must be fulfilled to meet the requirements of the project.

Each activity in a WBS is called Work Package (WP). Each WP should represent an identifiable work product traceable to one or more system level requirements.

In this section, the WBS for Blue Horizon Mission is presented and a brief description is given.

The mission will be divided into four principal WPs: Management, Space segment, Launch and Systems Engineering as it can be seen in Figure 3.1.

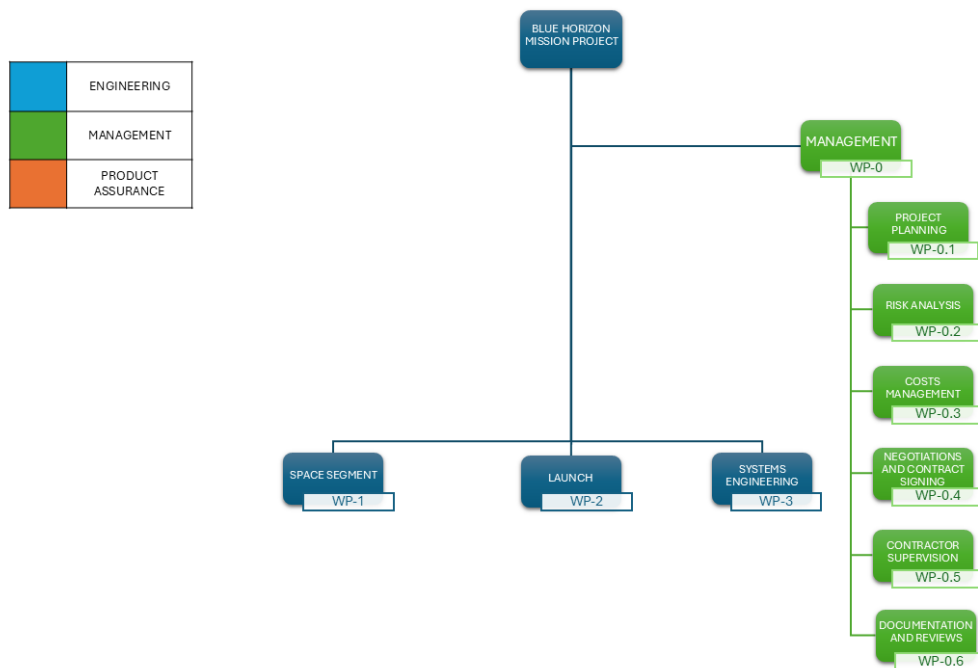


Figure 3.1: Work Packages of Blue Horizon Mission.

The space segment is divided into Platform, Payload & Models WPs as shown in Figure 3.2.

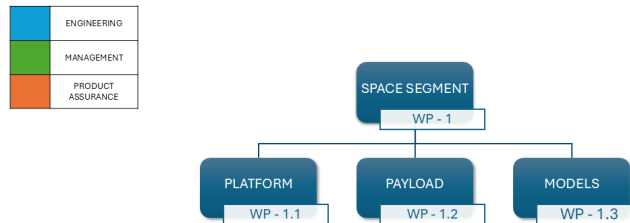


Figure 3.2: Space segment work packages

The platform WP consists on the seven different subsystems, including Structure, Thermal Control, Propulsion, Communications & Data Handling, Attitude Determination & Control and Mission Analysis as represented in the Figure 3.3. This WP includes the different requirements of each subsystem as well as the selection of components and the reviews and documentation provided for each subsystem.

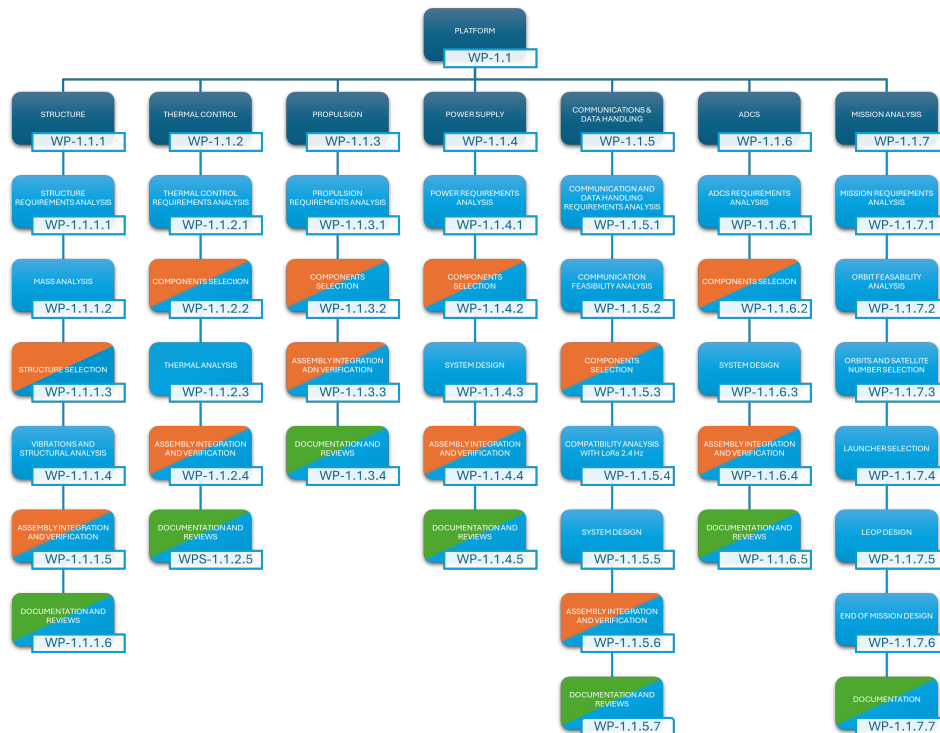


Figure 3.3: Platform work packages

As can be seen in Figure 3.4, the payload WP is divided into Payload Selection, Assembly, Integration and Verification and finally a third WP related to the documentation and the reviews which can be done during the development of the mission.

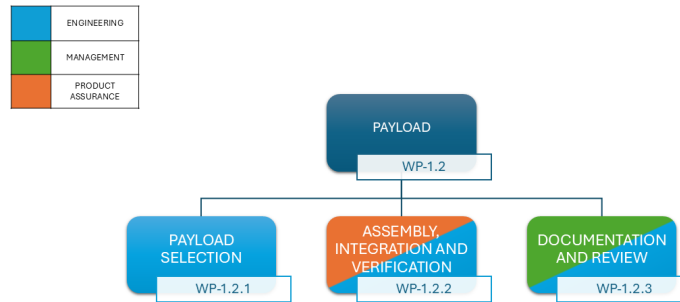


Figure 3.4: Payload work packages

The models WP contains, as it is shown in Figure 3.5, the models and the tests to be done during the development of the mission, it is, qualification model assembly and qualification test of each model, acceptance model assembly and integration, acceptance test and the part related with the documentation.

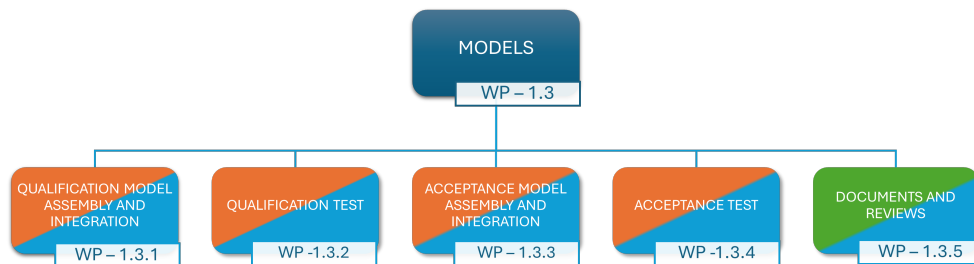


Figure 3.5: Models work packages

The Launch WP, Figure 3.6, includes every WP related with the launch, including date and costs analysis, negotiations, the launch campaign preparation including, the satellites transportation and integration and the operations, as well as the documentation and review documents of each WP.

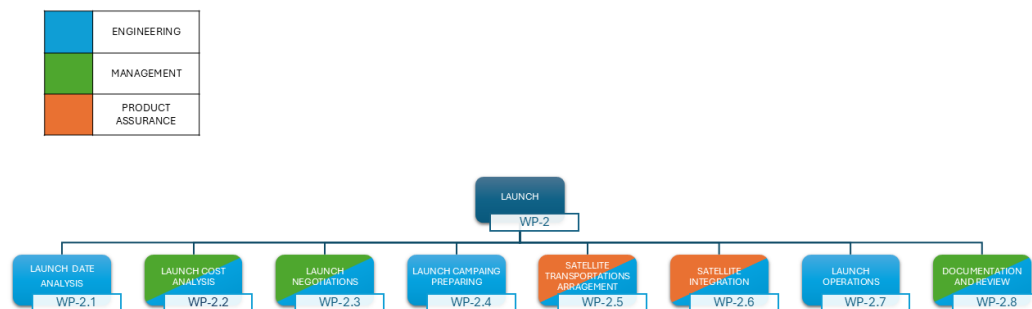


Figure 3.6: Launch work packages

The last part of the WBS, Figure 3.7, includes the functions related to the work of systems engineers. Including requirements definition and planning of the mission, system design and architecture (in this WP the selection of the technologies is included), coordination and supervision of each subsystem, operational readiness and monitoring and the corresponding part of project management and documentation.

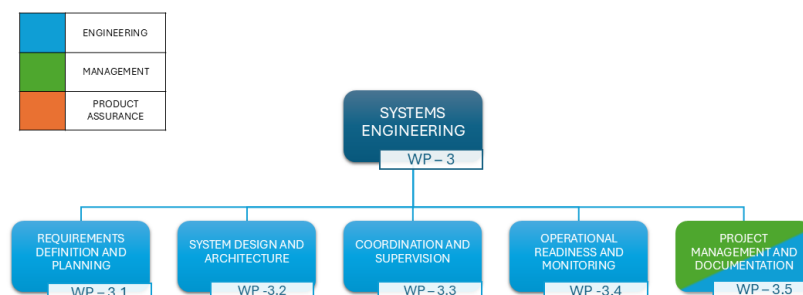


Figure 3.7: Systems Engineering work packages